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A-69

SC

SARDAR PATEL UNIVERSITY
B.C.A. (SEMESTER – II) EXAMINATION
Monday, 28th March 2016
2.30 p.m. to 4.30 p.m.
MATHEMATICS-II (US02FBCA02)

No. of printed pages: 04

Total Marks: 70

Q.1 Fill in the blanks:

10

1. $C(12, 2) =$ _____
 (a) 66 (b) 10 (c) 120 (d) 12
2. The vertex is also called _____
 (a) node (b) dot (c) full stop (d) one
3. A graph with no vertices and no edges is called _____
 (a) Trivial graph (b) multigraph (c) null graph (d) none
4. The Euler's formula is _____
 (a) $R + E - V = 2$ (b) $V - E - R = 2$ (c) $E - R + V = 2$ (d) $V - E + R = 2$
5. Four persons out nine persons can be selected in _____ ways.
 (a) $\binom{9}{5}$ (b) $\binom{9}{4}$ (c) $\frac{9!}{4!}$ (d) 24
6. The number of edges in the complete graph K_5 is _____
 (a) 15 (b) 25 (c) 10 (d) 5
7. $0! + 1! =$ _____
 (a) 0 (b) 1 (c) 2 (d) None of these
8. The second quartile Q_2 coincides with _____
 (a) Mean (b) Mode (c) Median (d) Standard deviation
9. Range of the Bowley's coefficient of Skewness is _____
 (a) -1 to 0 (b) -1 to 1 (c) 0 to 1 (d) -3 to 3
10. Quartile deviation is defined as _____
 (a) $\frac{Q_3 - Q_1}{2}$ (b) $\frac{Q_3 - Q_1}{3}$ (c) $\frac{Q_1 - Q_2}{2}$ (d) $\frac{Q_3 + Q_1}{2}$

Q.2 Attempt any ten.

20

1. Draw: K_5 & $K_{3,4}$
2. Draw a diagram for each of a graph $G=G(V, E)$:
 $V = \{A, B, C, D\}, E = [\{A, A\}, \{B, C\}, \{D, A\}, \{B, A\}, \{C, A\}, \{A, D\}]$
3. Define the degree of a vertex with example.
4. Draw the planar representation for the given graph $\rightarrow$



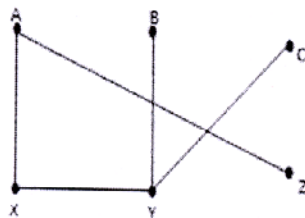
5. Prove that $\begin{pmatrix} 17 \\ 6 \end{pmatrix} = \begin{pmatrix} 16 \\ 5 \end{pmatrix} + \begin{pmatrix} 16 \\ 6 \end{pmatrix}$
6. What do you mean by the chromatic number of the graph? Find the chromatic number of K_{10} .
7. There are 5 bus lines between A and B and there are 4 bus lines between B and C.
Find the number of ways that a man can travel by bus : (1) From A to C by way of B
(2) Round trip from A to C by way of B.
8. Find the number of distinct permutation that can be formed from all the letters of the word (1) DADDY (2) THIS
9. Draw two connected maps of a graph with 6 edges and 3 regions.
10. If the quartiles $Q_1 = 62$, $Q_2 = 142$ and $Q_3 = 195$, then find the Bowley's coefficient of skewness.
11. Define Range and Quartile deviation.
12. Find standard deviation of first 5 natural numbers.

Q.3

- (a) Draw the graph G corresponding to each adjacency matrix given below. 05

$$(1) A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 1 & 3 & 0 \end{bmatrix} \quad (2) B = \begin{bmatrix} 0 & 2 & 1 & 1 \\ 2 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- (b) Define bridge and cut points. Find bridges and cut points for the following Graph. 05

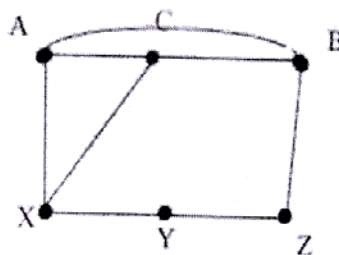


Q.3

OR

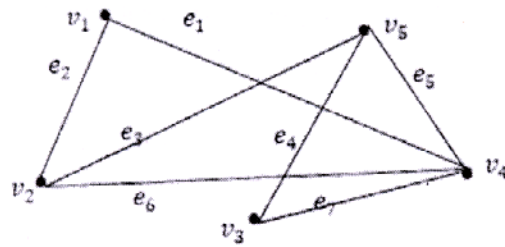
- (a) Consider the graph:-→

- (1) Describe the graph formally.
(2) Find the degree of each vertex.
(3) Verify that the sum of the degrees of the Vertices of a graph is equal to twice the Number of edges.



- (b) Find the adjacency & Incidence matrix of the graph given below:

05



Q.4

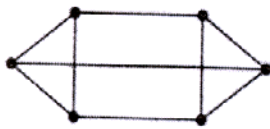
- (a) Find the rest using Euler's formula for the following connected planar multigraphs having vertices V , edges E and regions R :

04

(1) $V = 10, E = 14$ (2) $E = 10, R = 8$

- (b) Check whether the following graphs are planar or not. Justify your answer.

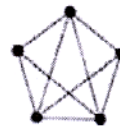
06



(1)



(2)



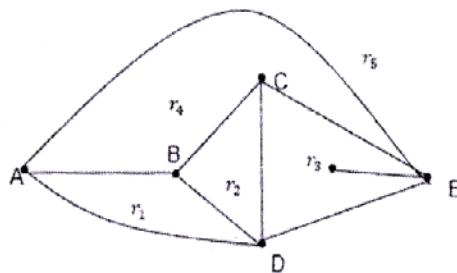
(3)

Q.4

OR

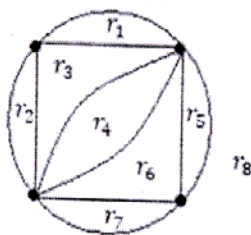
- (a) Identify cycle or closed path that borders each region of the following map. Also find the degree of each region.

05

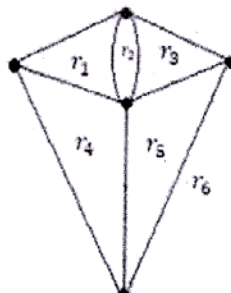


- (b) Paint the following maps with minimum number of colors:

05



(1)



(2)

Q.5

- (a) Find the number n of distinct permutations that can be formed from all the letters of the following words 03

1. MISSISSIPPI
2. AMARNAGAR
3. COMPUTER

- (b) Simplify: (1) $\frac{(n+1)!}{n!}$ (2) $\frac{(n+1)!}{(n-1)!}$ 03

- (c) Find n : (1) $P(n, 2) = 72$ (2) $P(n, 4) = 42P(n, 2)$ 04

Q.5**OR**

- (a) A farmer buys 3 cows, 2 pigs and 4 hens from a man who has 7 cows, 5 pigs and 9 hens. How many choices does the farmer have? 04

- (b) Find the number of three letter words using the six letters A, B, C, D, E and F without repetition. 03

- (c) How many committees of five with a given chair person can be selected from twelve persons? 03

Q.6

- (a) Calculate Karl Pearson's coefficient of correlation between x and y from the following data: 06

X:	10	6	9	10	12	13	11	9
Y:	9	4	6	9	11	13	8	4

- (b) Write types of correlation. Also explain positive correlation. 04

OR**Q.6**

- (a) Find quartile deviation for the following data: 05

Class	0-15	15-30	30-45	45-60	60-75	75-90	90-105
f	8	26	30	45	20	17	4

- (b) Following table shows the frequency distribution of marks. 05

Marks :	0-10	10-20	20-30	30-40	40-50	50-60
No. of students:	12	18	27	20	17	6

Find the Mean deviation from mean for above frequency distribution.
